

Curated Research Articles

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- **Decoding the Low-Voltage Plateau in Hard Carbon Negative Electrodes: From Sodium-Storage Models to Quantitative Active-Site Design** — score: 1.000 This review article investigates the complex sodium storage mechanisms of hard carbon anodes used in sodium-ion batteries, addressing the interplay of intercalation, pore filling, and active-site contributions. By reevaluating existing sodium-storage models and emphasizing the significance of microstructural features, the authors propose a new mechanistic framework aimed at enhancing the design of high-performance anodes through quantitative active-site optimization and improved material characterization techniques. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Copper-Governed Divergence of Lithium Deposition Modes in Anode-Free Lithium Metal Batteries** — score: 1.000 The study reveals that the microstructure of copper current collectors significantly influences lithium deposition modes in anode-free lithium metal batteries. Two distinct modes are identified: the localized nucleation-selective growth (LS) which results in poor-quality Li deposits, and the distributed nucleation-competitive coalescence (DCC) that produces stable, continuous layers. The findings suggest that optimizing the copper structure is crucial for improving battery performance and longevity, beyond traditional electrolyte and interphase engineering approaches. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Glycine-Driven Dynamic Interfacial pH Buffering Enables Highly Reversible Zinc Metal Anodes** — score: 1.000 The article discusses the use of glycine as an electrolyte additive in aqueous zinc-ion batteries to improve performance by regulating local pH and enhancing the chemistry at the zinc anode interface. Glycine's reversible protonation-deprotonation behavior effectively mitigates side reactions like dendrite growth and by-product formation, resulting in significantly increased cycling stability and efficiency in battery operation. This research highlights the importance of interfacial pH management in the development of advanced battery electrolytes. Journal: *Wiley: Small Methods: Table of Contents*
- **Process-Oriented Interphase Regulation in Lithium Metal Batteries with Non-Aqueous Liquid Electrolytes** — score: 1.000 The article presents a comprehensive review on the interphase regulation of lithium metal batteries (LMBs) using a six-step kinetic framework that encompasses key processes such as ion transport and nucleation. It explores the impact of various strategies, including electrolyte additives and artificial interphases, on enhancing interfacial stability and promoting uniform lithium deposition while addressing challenges related to dendrite formation and capacity decay, ultimately aiming to advance the practical use of LMBs. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Structure of Li₃YCl₆ and Its Effect on Lithium Ionic Conductivity** — score: 1.000 The study investigates the structural characteristics of Li₃YCl₆ and their impact on lithium ion conductivity, revealing that the distribution of Y-vacancies within the material's hexagonal close-packed structure significantly influences Li⁺ diffusion behavior. Lower temperature synthesis leads to a more uniform cation and vacancy distribution, enhancing conductivity, while high temperatures promote phase segregation that impairs mobility. Using kinetic Monte Carlo simulations, the research clarifies the structural role of Y-vacancies and addresses previous debates on conductivity optimization in halide-based electrolytes. Journal: *Journal of the American Chemical Society advanceAccess*
- **Revealing Non-Steady-State Structure-Performance Relationship in Plateau-Capacity-Dominated Hard Carbon Anodes for High-Rate Reversible Sodium Storage** — score: 1.000 The article highlights how the typical structure-performance relationship for hard carbon anodes in sodium storage does not hold true at varying temperatures. While these anodes show poor rate performance at room temperature, they significantly improve under elevated temperatures due to enhanced ion/electron transport and a beneficial solid electrolyte interphase, which together enable better sodium storage. This finding establishes a dynamic, non-steady-state relationship that offers new guidelines for optimizing sodium ion batteries under practical conditions. Journal: *Journal of the*

- **Synergistic Integration of Hierarchical Secondary Si/C Microrod Architecture and Fluorinated Ether Electrolyte for Stable Lithium-Ion Batteries** — score: 1.000 The article presents an innovative approach for enhancing lithium-ion battery performance by integrating a hierarchical secondary Si/C microrod architecture with a tailored fluorinated ether electrolyte. This combination effectively addresses silicon’s volume expansion and mechanical issues, yielding significant improvements in capacity retention and electrochemical stability, as demonstrated in extensive cycling tests. The findings suggest that this synergy offers a promising pathway for the development of more efficient and durable Si-based anodes in next-generation batteries. Journal: *ACS Applied Energy Materials advanceAccess*
- **Screening Lithium-Ion Conductors Based on Local and Global Topology** — score: 1.000 This article presents a novel hierarchical screening framework for identifying high-performance lithium-ion conductors, utilizing local structure order parameters and persistent homology to enhance the prediction of ionic conductivity. The study successfully screened 3,664 compounds, identifying 778 potential candidates, and validated the amorphous form of $\text{Li}_5\text{SbS}_3\text{I}_2$ as a solid-state electrolyte that significantly improved ionic conductivity through mechanical amorphization. The findings highlight the importance of specific local and global migration pathways in designing advanced all-solid-state batteries. Journal: *Journal of the American Chemical Society advanceAccess*
- **Interplay of Lattice Distortion and Cation Disorder Governs Li-Ion Transport in Cation-Disordered Rocksalt Cathodes** — score: 1.000 The study reveals that lattice distortion in cation-disordered rocksalt cathodes significantly affects lithium ion (Li^+) transport by reshaping percolation networks, contrary to traditional models that assume ideal lattice conditions. By integrating advanced simulations, the researchers demonstrate how enhanced distortions enable Li^+ migration through additional pathways, leading to the successful design of a new multication cathode with improved ion transport and electrochemical performance. Journal: *Journal of the American Chemical Society advanceAccess*
- **Coupled Evolution of Charged-State Stacking and Redox Partitioning in O3-Type Sodium Cathodes** — score: 1.000 The study explores the structural changes that occur in O3-type sodium cathodes at different charge voltages, specifically focusing on $\text{NaNi}_{1/3}\text{Fe}_{1/3}\text{Mn}_{1/3}\text{O}_2$. Charging to 4.2 V enhances capacity but leads to structural instability and accelerated degradation due to intergrowth with O-type slabs, while a 4.0 V cutoff promotes a more stable P3 framework, indicating a complex relationship between charged-state stacking, redox processes, and ion transport kinetics. Journal: *ACS Energy Letters advanceAccess*
- **Activating Aqueous Multivalent Metal–Sulfur Electrochemistry via Kinetic Promoter** — score: 1.000 The article discusses advancements in aqueous multivalent metal-sulfur batteries that enhance sulfur conversion kinetics through a newly introduced kinetic promoter. By employing elements like Li^+ and Mn^{2+} , the study reveals a mechanism that promotes solid-state reactions while suppressing undesirable dissolution, resulting in a highly efficient reversible sulfur redox chemistry. The batteries demonstrated impressive capacities and performance, suggesting this approach could be applicable to other multivalent aqueous systems. Journal: *Journal of the American Chemical Society advanceAccess*
- **Li P F O Bonding Topology Governs Oxygen Redox and Interfacial Stability in High-Rate LRMO-Based All-Solid-State Lithium Batteries** — score: 1.000 The study presents a novel Li–P–F–O bonding topology that enhances the stability of surface oxygen in lithium-rich manganese-based oxide cathodes for all-solid-state lithium batteries. By employing mechanochemical fluorophosphorylation, this approach mitigates oxygen release and interfacial degradation, leading to improved cycling performance and capacity retention, demonstrating the potential of bonding topology regulation in optimizing battery efficiency. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Recent Progress Toward Practical Aqueous Zn-I₂ Batteries: From Lab to Practical** — score: 0.900 This review explores advancements in aqueous Zn-I₂ batteries, highlighting the energy-

storage mechanisms and challenges such as polyiodide dissolution and dendrite growth. It details various strategies, including chemical and physical approaches, developed to enhance efficiency and stability, particularly under high iodine loading and pouch cell conditions, aiming for more practical applications in energy storage. Journal: *Wiley: Advanced Energy Materials: Table of Contents*

- **A Critical Study of the Electrochemical Performance of MXenes for Energy Storage** — score: 0.900 This study systematically compares the electrochemical performance of various MXenes, including Ti₃C₂Tx and Nb₂CTx, synthesized via a fluoride-salt etching method. The findings illustrate that structural stoichiometry, surface chemistry, and morphology critically influence their charge-storage characteristics and stability in sulfuric acid, revealing specific capacitances and stability limits that vary among the different MXene phases. Journal: *ACS Applied Energy Materials advanceAccess*
- **Configurational-Entropy-Engineered Prussian Blue Analogue Cathodes for Low-Temperature Semi-solid-state H₂ Batteries** — score: 0.900 The study introduces a novel cathode material, CEM . -PBA, engineered with configurational entropy to enhance the performance of semi-solid-state H₂ batteries at low temperatures. The findings show significant improvements in proton transport kinetics and storage capabilities, allowing the battery to operate effectively at temperatures as low as -60 °C, with excellent cycling stability and capacity retention over thousands of cycles. Journal: *ACS Energy Letters advanceAccess*
- **Doping-controlled silicon interlayers steer solid-electrolyte interphase formation in lithium metal batteries** — score: 0.800 The article discusses how doping silicon interlayers can influence the formation of the solid-electrolyte interphase in lithium metal batteries. This approach potentially enhances battery performance by optimizing the chemical interactions at the interface, leading to improved stability and efficiency in energy storage applications. Journal: *ScienceDirect Publication: Energy Storage Materials*
- **Advanced Interfacial Characterization Techniques for Aqueous Zinc-ion Batteries** — score: 0.800 The article discusses cutting-edge techniques for analyzing interfacial interactions in aqueous zinc-ion batteries. By improving characterization methods, the authors aim to enhance the understanding of these battery systems, potentially leading to advancements in their efficiency and longevity. Journal: *ScienceDirect Publication: Nano Energy*
- **Dense Zinc Deposition Underneath a Zincophobic MoTe₂ Protective Layer via Weakening Electrostatic Repulsion for Durable Zinc Metal Batteries** — score: 0.800 The article discusses the development of a zincophobic MoTe₂ protective layer on zinc foil anodes for aqueous zinc metal batteries. This layer establishes an interfacial electric field that weakens electrostatic repulsion and compresses the electrical double layer, facilitating dense, dendrite-free zinc deposition and significantly enhancing the battery's cycle life and performance, reaching up to 5000 hours of operation. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Optical Fingerprinting of Rate-Dependent Staging Dynamics in Individual Graphite Particles** — score: 0.800 The article explores the optical fingerprinting technique to investigate the rate-dependent staging dynamics of individual graphite particles. It highlights the importance of these dynamics in understanding energy storage materials, particularly in enhancing the efficiency and performance of battery technologies. Journal: *ScienceDirect Publication: Energy Storage Materials*
- **Hydrogen-Bond-Regulated and Rheologically Adaptive Interphase for Corrosion-Suppressed Dense Zinc Deposition in Long-Lasting 4 Ah Zinc Metal Batteries** — score: 0.800 The study presents a novel interphase design for zinc metal batteries that utilizes a hydrogen-bond-regulated ionic liquid layer and a rheologically adaptive polydimethylsiloxane gel. This innovative interphase mitigates water-induced corrosion and promotes stable zinc deposition by accommodating interfacial stress, leading to significantly improved cycling stability and capacity in high-current conditions for aqueous zinc batteries. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Single-Source Lithium Phosphate Precursor Engineering for Sustainable and Supply-Chain-Resilient Lithium Iron Phosphate Cathodes** — score: 0.800 The article presents a novel synthesis method for lithium iron phosphate (LiFePO₄) cathodes using lithium phosphate as a

single-source precursor, which enhances the material’s electrochemical performance and sustainability by reducing environmental impact and supply chain risks associated with traditional multi-precursor systems. The resulting cathode exhibits impressive capacity and retention in battery cycling, demonstrating a shift towards greener production methods and compatibility with recycling efforts. Journal: *Wiley: Small Methods: Table of Contents*

- **Defect-Targeted Passivation Enabling Capacity Compensation of Hard Carbon Anodes for Sodium-Ion Batteries** — score: 0.800 The article presents a novel defect-targeted passivation approach to enhance the performance of hard carbon anodes in sodium-ion batteries, addressing issues of rapid capacity decay and poor rate capability. By selectively passivating reactive edge defects and leveraging topological defects for reversible sodium storage, the optimized anodes achieve impressive metrics, including a reversible capacity of 405.7 mAh g⁻¹ and stable operation across a wide temperature range, thus advancing the efficiency and durability of sodium-ion battery technology. Journal: *ACS Energy Letters advanceAccess*
- **Regulating Cation-Solvent Coordination in Carbonate Electrolytes for Low-Temperature and Acid-Resistant Sodium-Ion Batteries** — score: 0.800 This study introduces vinyltrimethylsilane (VTMS) as an additive in carbonate electrolytes for sodium-ion batteries, addressing the instability of hard carbon anodes. By improving the electrolyte phase and interfacial layers through VTMS’s action, the resulting batteries show enhanced rate capability and cycling stability, achieving impressive performance metrics even at low temperatures and demonstrating a valuable strategy for electrolyte refinement in energy storage systems. Journal: *ACS Energy Letters advanceAccess*
- **Pressure-Driven Performance and Thermal Safety Tradeoff in Solid-State Batteries** — score: 0.800 The study investigates the effects of external pressure on the performance and thermal safety of solid-state batteries (SSBs). It finds that while increased pressure enhances electrochemical capacity and thermal efficiency, it also creates complex thermal responses that must be managed to mitigate risks of thermal runaway. The findings provide guidelines for optimizing SSB designs to balance performance and safety in high-power applications. Journal: *ACS Energy Letters advanceAccess*
- **Chemical Bonding Interface Enables Lithium Batteries to Operate Under Near-Vacuum Conditions** — score: 0.800 The article presents a novel Li O chemical bonding interface prototype that allows lithium metal batteries to function efficiently under near-vacuum conditions, eliminating the need for high stack pressure. This interface enhances lithium ion transport and adhesion strength, resulting in high capacity retention and coulombic efficiency in battery cells, which could significantly benefit lightweight designs for aerospace applications. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Highly Efficient and Cost-Effective Single-Atom Catalysts for Lithium-Sulfur Batteries Exceeding 540 Wh kg⁻¹** — score: 0.700 The study presents an innovative manganese single-atom catalyst anchored on carbon nitride, designed to enhance the performance of lithium-sulfur batteries. By utilizing a unique dual-coordination mechanism, the catalyst significantly improves polysulfide conversion kinetics, leading to pouch cells with an impressive energy density of 547 Wh kg⁻¹ and stable cycling, thus advancing the practical deployment of high-energy-density lithium-sulfur batteries. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Rational Construction of a Dissolution-Resistant SEI via Competitive Reduction Mechanism for Stable Sodium Metal Batteries** — score: 0.700 This article discusses a novel approach to enhance the stability of sodium metal batteries by developing a robust solid-electrolyte interphase (SEI) through a competitive reduction mechanism. The authors present a strategy aimed at minimizing dissolution issues, ultimately improving the performance and longevity of sodium metal batteries. Journal: *ScienceDirect Publication: Energy Storage Materials*
- **Orbital Hybridization Engineering Stabilizes Sulfate Cathodes by Strengthening Fe–O Covalency and Suppressing Fe Migration** — score: 0.700 The study presents a novel approach to stabilize sulfate cathodes in energy storage systems by engineering orbital hybridization, which enhances the covalency of the Fe–O bonds. This modification effectively suppresses Fe migration, po-

tentially improving the performance and longevity of these cathodes in practical applications. Journal: *ScienceDirect Publication: Energy Storage Materials*

- **Iron-Based Sodium-Salt Composite Cathode for Sodium-Ion Batteries (Adv. Energy Mater. 35/2026)** — score: 0.600 The article presents a novel sodium-salt composite cathode for sodium-ion batteries, developed by one-step ball milling Fe and Na₂SO₄ with CuS. The unique structure, enhanced by amorphous CuS, facilitates charge transfer and Na-ion transport, enabling efficient multielectron Fe redox reactions and promoting sustainability in battery technology. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Electric Double Layer Reconstruction via Dielectric-Response-Driven Dynamic Adsorption Enables Ultralow-Overpotential Long-Life Zn Anodes** — score: 0.600 The article discusses a novel method for enhancing the performance of zinc anodes in energy storage applications through dynamic adsorption processes that drive electric double layer reconstruction. This approach results in ultralow overpotential and prolonged lifespan for the anodes, highlighting its potential for improving energy storage efficiency. Journal: *ScienceDirect Publication: Energy Storage Materials*
- **Chaotropic Urea-Regulated PAM-Agar Hydrogel Electrolytes for Flexible Zn-Air Batteries** — score: 0.600 The article presents a novel strategy for enhancing the ionic conductivity of dual-network PAM-agar hydrogels used in flexible Zn-air batteries by incorporating urea as a chaotropic agent. This addition improves the hydrogel's structure and ion transport, leading to significant performance gains in battery metrics such as open-circuit voltage and stability across multiple charge-discharge cycles. Journal: *Energy & Fuels advanceAccess*
- **A Physics-Guided Generative Diffusion Model for Super-Resolution AFM of Trap Dynamic Behavior** — score: 0.500 The article introduces StableAFM, a novel generative model that enhances the speed and resolution of atomic force microscopy (AFM) imaging by effectively combining a physically-informed degradation model with latent-space diffusion sampling. This method significantly improves the reconstruction of high-fidelity images from sparse data and successfully accelerates Kelvin probe force microscopy imaging, revealing complex charge dynamics in multilayer ceramic capacitors, including previously unobserved depletion zones. Journal: *Wiley: Small Methods: Table of Contents*
- **Asymmetrically Supported Electrolytes in Organic Redox Flow Batteries: A New Strategy for Improving Performance and Unexpected Effects Due to Trace Oxygen Ingress** — score: 0.400 The article discusses a novel approach to enhancing the performance of organic redox flow batteries through the use of asymmetrically supported electrolytes. Additionally, it explores the unexpected effects that can arise from trace amounts of oxygen entering the system, which may influence battery efficiency and longevity. Journal: *ScienceDirect Publication: Energy Storage Materials*
- **Two-Dimensional Orthorhombic Molybdenum Boride Enabled by Selective Mo–Al Bond Cleavage in MoAlB** — score: 0.400 The article discusses a novel method for producing two-dimensional (2D) orthorhombic molybdenum boride (MoB) nanosheets from a non-van der Waals MoAlB precursor using selective bond cleavage facilitated by liquid metals. This technique effectively prevents unwanted structural changes and leads to the creation of layered MoB with enhanced catalytic properties, ultimately improving the performance and stability of anode-free lithium metal batteries. Journal: *Journal of the American Chemical Society advanceAccess*
- **In Situ Polarization Imaging of Multi-Bubble Dynamics on Nickel-Based Electrodes During Hydrogen Evolution** — score: 0.300 The study presents a novel dual-camera polarization imaging technique that allows for real-time analysis of multi-bubble dynamics during the hydrogen evolution reaction on nickel-based electrodes. By introducing a new apparent coverage parameter, DOP, it quantifies the interfacial state without needing to pinpoint individual bubble boundaries, revealing insights into bubble behavior influenced by different electrochemical conditions, which enhances the understanding of catalytic performance in electrolysis systems. Journal: *Wiley: Small Methods: Table of Contents*
- **Sequentially segment-based identification of key parameters and effective reconstruction**

- of electrochemical impedance spectroscopy for lithium-ion batteries** — score: 0.300 The article discusses a method for enhancing the analysis of electrochemical impedance spectroscopy (EIS) in lithium-ion batteries by employing a sequential segmentation approach. This technique aims to identify critical parameters and improve the accuracy of EIS data reconstruction, potentially leading to better battery performance assessments. Journal: *ScienceDirect Publication: Energy Storage Materials*
- **Mechanical signals decode Li-deposition porosity for failure analysis and early warning in practical lithium metal batteries** — score: 0.300 The article discusses a novel approach for analyzing and predicting failures in lithium metal batteries by examining mechanical signals that decode porosity related to lithium deposition. This technique aims to enhance early warning systems, thereby improving the reliability and safety of these energy storage devices. Journal: *ScienceDirect Publication: Energy Storage Materials*
 - **Materials design and integration strategies for next-generation smart batteries** — score: 0.300 The article examines essential materials and integration approaches for advanced smart batteries that can monitor their health, adapt to internal conditions, and facilitate informed management. It highlights the importance of responsive materials and innovative sensing technologies to create sustainable, cell-compatible battery systems. Journal: *Nature Reviews Materials*
 - **Screening and Experimental Validation of Bifunctional Organic Lithium Salts for Direct Regeneration of Spent LiFePO₄ Cathodes** — score: 0.200 The article discusses the identification and testing of bifunctional organic lithium salts designed to regenerate spent LiFePO₄ cathodes directly. The authors detail their screening process and experimental validation, highlighting advances in improving the sustainability and performance of lithium-ion batteries. Journal: *ScienceDirect Publication: Energy Storage Materials*
 - **Interfacial Engineering of Intrinsic and Extrinsic Interfaces in All-Solid-State Batteries** — score: 0.200 The article discusses advancements in interfacial engineering within all-solid-state batteries, focusing on both intrinsic and extrinsic interfaces. It aims to enhance battery performance and longevity by optimizing these interfaces, which are crucial for effective ion transport and overall device efficiency. Journal: *ScienceDirect Publication: Nano Energy*
 - **A Universal Synthesis of PtPd-Based High-Entropy Sub-Nanoribbons for Oxygen Reduction Electrocatalysis** — score: 0.200 This article presents a novel synthesis method for producing PtPd-based sub-nanoribbons (SNRs) with tunable configurational entropy, enhancing their effectiveness in catalyzing the oxygen reduction reaction (ORR). Testing indicates that higher entropy correlates with improved catalytic performance, ultimately leading to the development of the high-entropy Pt-PdNiFeGa SNRs, which demonstrate exceptional mass activity and stability, surpassing many existing PtPd-based catalysts. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
 - **Emerging High-Entropy Materials for Rechargeable Lithium Batteries** — score: 0.200 The article discusses the potential of high-entropy materials in enhancing the performance of rechargeable lithium batteries. It highlights recent advancements and the role these materials play in improving energy storage capabilities and battery longevity. Journal: *ScienceDirect Publication: Energy Storage Materials*
 - **IDART: Interferometric Dual-AC Resonance Tracking for Nano-Electromechanical Mapping** — score: 0.200 The article presents interferometric dual-AC resonance tracking (iDART), an advanced imaging technique that enhances piezoresponse force microscopy (PFM) by significantly improving sensitivity and reducing bias artifacts. By achieving femtometer-level resolution and a tenfold increase in signal-to-noise ratio, iDART enables precise imaging of ferroelectric domains and reliable hysteresis loop characterization at low electrical biases, thereby expanding the capabilities of nanoscale electromechanical mapping in various materials, including hafnia-based ferroelectrics. Journal: *Wiley: Small Methods: Table of Contents*
 - **Femtosecond X-ray Tracking of Oxygen Vacancy-Driven Shallow-to-Deep Trap Transition and Its Impact on Charge Carrier Dynamics in Blue TiO₂** — score: 0.200 The study employs femtosecond X-ray transient absorption spectroscopy to investigate how oxygen vacancies in blue TiO₂

alter the charge carrier dynamics. It finds that these vacancies lead to a sequential cooling of hot electrons into shallow traps followed by a transition to deeper traps, which significantly slows down charge recombination compared to pristine TiO₂, thereby enhancing photocatalytic efficiency. The research provides a detailed understanding of the relationship between oxygen vacancy density and charge carrier behavior in TiO₂. Journal: *Journal of the American Chemical Society advanceAccess*

- **Rate Coefficients by Integration of Total Reaction NMR Spectra** — score: 0.200 The article discusses a novel approach for estimating reaction rate coefficients through the integration of total reaction NMR spectra and cumulative temporal concentrations. By analyzing six case studies involving various chemical processes, the authors demonstrate that this method simplifies the determination of kinetic parameters without the need for complex curve fitting techniques, thus enhancing the understanding of reaction mechanisms. Journal: *Journal of the American Chemical Society advanceAccess*
- **Ultrathin composite polymer electrolyte via in-situ polymerized poly(vinylene carbonate) within an electrospun scaffold for high energy lithium metal batteries** — score: 0.200 The article discusses the development of an ultrathin composite polymer electrolyte made through in-situ polymerization of poly(vinylene carbonate) within an electrospun scaffold, which aims to enhance the energy efficiency of lithium metal batteries. This innovative electrolyte design could significantly improve battery performance and longevity. Journal: *ScienceDirect Publication: Energy Storage Materials*
- **All-In-One Sensing Supercapacitor With Shared Configuration and Decoupled Performance** — score: 0.100 The article presents an innovative all-in-one sensing supercapacitor (ASS) designed for simultaneous strain sensing and energy storage, enhancing wearable technology integration. It features a shared configuration that facilitates efficient ion transport and robust electron flow, achieving impressive electrochemical performance, including high capacitance and stability under compression. The ASS demonstrates excellent sensing capabilities and minimal interference during operation, supporting its potential for diverse wearable applications. Journal: *Wiley: Advanced Energy Materials: Table of Contents*
- **Encoded and non-genetic protein variants expand human functional proteome** — score: 0.000 The article discusses how both encoded and non-genetic variations in proteins contribute to an expanded functional proteome in humans. It highlights the importance of these variants in understanding human biology and disease, emphasizing their potential impact on proteomics and therapeutic development. Journal: *Nature*
- **The roots of astrology, the mysterious nature of time and more: Books in brief** — score: 0.000 In a recent review published in *Nature*, Andrew Robinson highlights five notable science books, exploring topics such as the origins of astrology and the enigmatic concept of time. The selections aim to offer readers insights into these intriguing subjects through accessible and thought-provoking narratives. Journal: *Nature*
- **How is climate change affecting conifers and their ability to sequester carbon?** — score: 0.000 The article discusses recent research that highlights the complex interplay of various factors influenced by climate change, which may hinder the growth of conifer trees and their capacity to sequester carbon effectively. This could have significant implications for carbon storage and forest ecosystems. Journal: *Nature*